Palaeomagnetism of the Karoo igneous rocks in southern Africa

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Virtual geomagnetic poles have been derived from over 2000 samples from 691 sites in Karoo igneous rocks, ranging from basalts and dolerites through rhyolites, and at localities spread from Namibia through the central Karoo basin, Lesotho to the Lebombo. After appropriate analysis, poles derived from Lesotho lavas, Karoo dolerite dykes and sills, Sabie River basalts, Mashikiri and Letaba lavas, and Jozini rhyolites, when averaged, yielded a palaeomagnetic (south) pole at: N = 6, latitude = –69.2, longitude = 98.3, α = 3.3. The sites in Lesotho come from thirteen sections sampled through the entire basalt section, the single reversal encountered being traced throughout this country. This appears to correlate both palaeomagnetically and geochemically with the major reversal found in the Sabie River basalt in all three of the sections sampled through the Lebombo stratigraphic pile. At least four reversals were found in the Lebombo, but paucity of outcrops and sampling precludes precise correlation. Together with new radiometric age data, the similarity of pole positions and the paucity of reversals are consistent with the hypothesis that all these Karoo magmatic products were emplaced between ~190 and ~175 Ma, with the bulk of the basalts erupted rapidly and penecontemporaneously throughout the country around 183 ± 2 Ma. The Mbuluzi rhyolites in Swaziland, which are stratigraphically younger than the Jozini rhyolites, yielded a pole: N = 9 sites, latitude = –45.6, longitude = 98.6, α = 10.2, which is distinct from that of the main Karoo magmatism, suggesting a significant difference in age. Four sites from the Bumbeni syenite complex in KwaZulu–Natal, dated at ~145 Ma, yielded bipolar results and a pole almost identical to that obtained previously from the coeval Main and Male kimberlite fissures at Swartruggens in the North West Province (also bipolar). The mean pole from these two studies: N = 6, latitude = –31.7, longitude = 104.3, α = 6.3, is considered to be another reasonably robust result for that vintage. Two sites in the Peninsula dolerites near Cape Town dated at 132 Ma, plus one from further east, yielded a mean VGP at latitude = –42.2, longitude = 105.2, α = 9.1.

Virtuele geomagnetiese pole is van oor die 2000 monsters uit 691 terreine op Karoo stollingsgesteentes, wat wissel van basalte en doloriete deur rioliete, en uit gebiede versprei van Namibië af deur die sentrale Karookom, Lesotho, en tot by die Lebomboberge vasgestel. Die pole wat van Lesotho-lawas, Karoodolerietgange en -plate, Sabierivierbasalte, Mashikiri- en Letabalawas, en Jozini-rioliet verkry is, het, nadat die toepaslike analise gedoen en die gemiddelde bepaal is, 'n paleomagnetiese (suid)-pool opgelewer by: N = 6, breedtegraad = –9.2, lengtegraad = 98.3, α = 3.3. Die terreine in Lesotho kom vanaf dertien seksies wat deur die totale basaltseksie bemonster is, en die enigste omkering wat teëgekom is, is dwarsdeur hierdie landstreek nagespoor. Dit kom voor asof hierdie bevinding paleomagneties asook geochemies met die hoofomkering gevind in die Sabierivierbasalt, en wel in al drie die seksies wat deur die Lebombo stratigrafiese kolom bemonster is, korreleer. Daar is ten minste vier omkerings in die Lebombogebied gevind, maar 'n skaarste aan dagsome en geneemde monsters verhinder presiese korrelasie. Nuwe radiometriese ouderdomsdata, die eendersheid van die poolposisies, en die seldsaamheid van omkerings is alles in ooreenstemming met die hipotese dat al hierdie magmatiese produkte van die Karoo Supergroep tussen ~190 en ~175 Ma ingeplaas is en dat die meerderheid van die basalte teen 183 ± 2 Ma vinnig en penekontemporêr dwarsdeur die landstreek uitgebars het. Die Mbuluzi-rioliete in Swaziland, wat stratigrafies jonger as die Jozini-rioliete is, lewer die volgende pool: N = 9 plekke, breedtegraad = –45.6, lengtegraad = 98.6, α = 10.2, wat 'n kontras vorm met dié van die hoof Karoo Magmatisme en dus op 'n beduidende ouderdomsverskil dui. Vier terreine van die Bumbenisiënietkompleks in KwaZulu–Natal, gedateer ~145 Ma, lewer bipolêre uitslae en 'n pool wat amper identies is aan dit wat voorheen van die kontemporêre Main- en Male-kimberlietsplete te Swartruggens in die provinsie Noordwes bepaal is (ook bipolêr). Die gemiddelde pool van hierdie twee studies: N = 6, breedtegraad = –31.7, lentegraad = 104.3, α = 6.3, word beskou as nog 'n redelike stewige uitslag vir daardie datering. Twee terreine in die Skiereilanddoleriet naby Kaapstad, waarvan die ouderdom op 132 Ma vasgestel is, plus nog een verder oos, lewer 'n gemiddelde VGP (virtuele geomagnetiese pool) van: Breedtegraad = –42.2, Lengtegraad = 105.2, α = 9.1.

Introduction

This paper reports the combined results of two complementary palaeomagnetic studies of the Karoo igneous rocks. Between 1985 and 1990 Hargraves collected approximately 500 oriented samples from 82 sites ranging geographically from the Cape Peninsula to the northeastern Transvaal. This sampling concentrated primarily on dolerite dykes (and some sills) throughout the Karoo basin, and the rhyolite lavas of the Lebombo monocline in KwaZulu–Natal, Swaziland, and eastern Northern Province (Figures 1 and 2). In 1993 and 1994 Hooper, Rehacek, Marsh, R.A. Duncan, and A.R. Duncan collected over 1500 samples from 589 sites; of these, 459 sites were collected from thirteen stratigraphic sections through the 1400-m-thick basalt pile in Lesotho (Figure 2). Others were collected from the more compositionally varied units (nephelinite, picrite, basalt, and rhyolite) in Kruger Park in the northern Lebombo region (Sweeney *et al.*, 1994); from dykes and sills in the Free State, Gauteng, and Mpumalanga, and in southern Namibia, from the Hardap section of the Kalkrand Basalt Formation and the Keetmanshoop dolerites (Figure 1). Aimed primarily at collecting fresh samples for dating and geochemical studies, the 1993 – 4 samples aver-

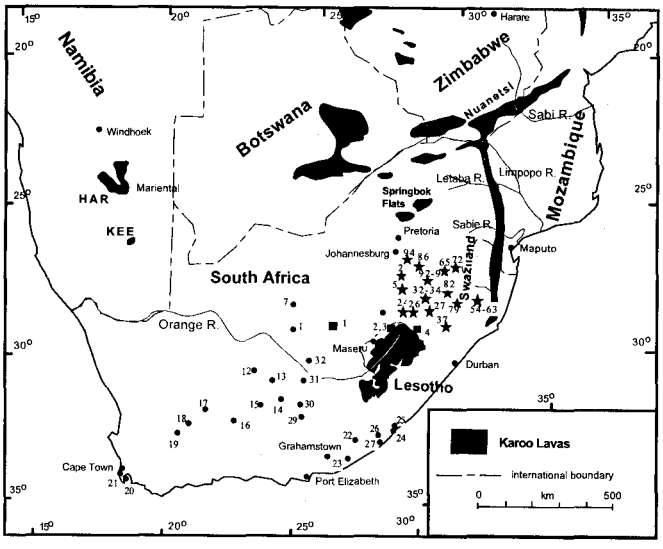


Figure 1 Location of samples discussed in this study. Dolerites: black dots (1985) and squares (1987), Hargraves; stars, Hooper and co-workers. Details of sampling sites in Lesotho and Lebombo are in Figure 2.

aged only two cores per site. In Lesotho this has enabled the magnetic reversal reported first by Van Zijl *et al.* (1962) near the base of the basalt flow succession to be traced over the whole of the country. A detailed geochemical study of the Sabie River basalts in the eastern Transvaal had been completed by Sweeney (1988) and Sweeney *et al.* (1994), and the 1993 – 4 oriented cores were collected from critical Sweeney sites (and elsewhere in the Lebombo) to determine the palaeomagnetism and the age of the magnetic polarity reversals recorded there.

Serendipitously, when these two independent studies are combined they constitute an intensive and geographically comprehensive palaeomagnetic and magnetostratigraphic restudy of the Karoo magmatic province across southern Africa. As Hooper and associates have also analysed chemically their own samples and all the dolerites sampled by Hargraves, this extensive coverage is applicable to their stratigraphic and geochemical studies which they will report separately.

Previous palaeomagnetic work

The wide distribution of the Karoo basaltic rocks made them the subject of some of the earliest palaeomagnetic studies undertaken in Africa (e.g. Graham & Hales, 1957; Graham *et al.*, 1962; Van Zijl *et al.*, 1962; see also Nairn, 1960). In the early work, alternating field (A.F.) demagnetization — if attempted at all — was rarely carried to very high fields and usually was applied in batch fashion. After systematic demagnetization of representative samples, all other samples would be demagnetized at what was judged to be the 'optimum' field (or fields), averages calculated, and reported with the usual Fisher statistics (Fisher, 1953). The early studies reported were mostly local and geographically scattered. Despite this, remarkably similar pole positions were usually obtained (see Table 1).

More recent (and experimentally robust) palaeomagnetic studies of Karoo igneous rocks were those of Gidskehaug *et al.* (1975) and Henthorn (1981) (Table 1). Gidskehaug *et al.*'s (1975) study in Namibia focused primarily on the Etendeka lavas which appear to correlate with the Serra Geral basalts in Brazil and are about 130 Ma, but three sites in the Hoachanas (now called Kalkrand) basalts were also sampled, along with sites in dolerite sills near Keetmanshoop. Samples from both of these latter localities yielded Karoo-like K–Ar ages, but the only useful magnetic results were obtained from the lavas (Table 1).

The presence of normally and reversely magnetized Karoo dykes and sills was first reported by Graham & Hales (1957) and was followed by the well-documented reversal in the Lesotho basalts by Van Zijl *et al* (1962). Henthorn (1981) reported a detailed palaeomagnetic study of the Sabie River basalts (possibly including Letaba picrites beneath) exposed along the Olifants River in the Kruger National Park (Table 1). A reversal from R to N in the lower part of the Sabie River basalts was found and it was suggested that this could be correlated with that observed in the Drakensberg by Van Zijl *et al.* (1962).

Table 1 Published Karoo paleomagnetic (South) poles and dates

Country	Unit	Sites	Age, Ma	Lat. S.	Long. E.	α	Ref
Namibia	Hoachanas* Lavas	3	161–173	62	72	16.9/23.1	1
Lesotho	Stormberg Lavas	74	193±3?	71	89	15	2
Lesotho	Lesotho Lavas	28	180±4	72	88	4.5	3
Zimbabwe	Marangudzi Complex	8	186±3	70	105	8.9	4
Zimbabwe	Shawa ijolite	5	204±16	64	86	14.1	5
South Africa	Sabie River basalts**	23	193±3?	68	95	8.6	6
Zimbabwe/Mozambique	Karoo lavas		154–190?	57	84	8	7
South Africa & Zimbabwe	Karoo dolerites	10	154–190?	65	75	12.3	8
Zimbabwe	Mateke Hills	6	173±4	59	80	8.5	4
South Africa	Swartruggens kimberlites	2 (12)	142±4	32	104	6.7	9

References: 1. Gidskehaug *et al.*, 1975; 2. Van Zijl *et al.*, 1962; 3. Kosterov & Perrin, 1996; 4. Brock, 1968; Foland & Henderson, 1976; 5. Gough & Brock, 1964; Nicolaysen *et al.*, 1962; 6. Henthorn, 1981; 7. McElhinny *et al.*, 1968; 8. Graham & Hales, 1957; McElhinny & Jones, 1965; 9. Hargraves, 1989; Allsopp *et al.*, 1989.
*Hoachanas lavas now called Kalkrand **Pole calculated from Henthorn's (1981) data

Table 2a $^{40}Ar/^{39}Ar$ dating at Princeton

Site*	N/n	$^{40}Ar/^{36}Ar$	Age, Ma
85-01	14/9	300±4	193.3±6.5
85-16	12/11	294±8	178.3±1.6
85-19	11/7	311±9	182.1±4.2
87-04	12/10	272±6	181.9±5.2
87-06	25/18	284±6	191.1±8.0
87-24 (Rooirand dyke)	17/13	376±6	200.8±4.5
87-9 (Bumbeni sanidine)	27/17	305±17	145.8±1.3

* For site locations, see Figures 1 and 2

N/n N = number of separate plagioclase grains fused with the YAG laser, n = number of data points used in the correlation diagram regression. Points with largest deviation from the regression line were removed until no real improvement in error occurred. No significant shift in age

$^{40}Ar/^{36}Ar$ = value extrapolated from correlation diagram

J-value = 0.01182±0.00008, based upon 520.4 Ma age for standard Mmhb-1 (Samson & Alexander, 1987)

Table 2b $^{40}Ar/^{39}Ar$ plateau ages* of Duncan *et al.* (1997)

Location	Samples measured	No. of samples	Age range, Ma
Lesotho	Lesotho basalts (+ plagioclase)	11	179.5–186.3
Lebombo	Jozini rhyolite	2	178.1–179.7
	Letaba picrite	1**	182.7
	Mashikiri nephelinite	1	182.1
	Sabie basalt (+ plagioclase)	4	181.2–184.2
Namibia	Hardap basalt (+ plagioclase)	3	183.0–186.0
	Keetmanshoop basalt (+ plagioclase)	4	180.5–184.7
Antarctica	Kinvan Mountains (plagioclase)	3	180.6–182.8

* Ages reported relative to biotite monitor FCT-3 (28.03±0.18 Ma) which is calibrated against an age of 523.5 Ma for Mmhb-1. ** One anomalously low at 141.9 Ma

Most recently, Kosterov & Perrin (1996) have reported the results of their study of four sections through the Lesotho lavas (see Table 1). Gratifyingly, their pole is not significantly different from the authors based on eleven sections through the same lavas (see below). Most intriguing, however, is that both of these poles are essentially indistinguishable from that for the same rocks published more than 30 years ago by Van Zijl *et al.* (1962).

Age of the Karoo magmatism

The Karoo Igneous Province was divided into a number of geographical areas by Eales *et al.* (1984), of which the central area includes the Lesotho basalts and the Karoo dolerites of the main Karoo basin. Throughout this central area, the composition of all but the earliest basalts is chemically uniform, and identical to most of the dykes and sills. It is on this basis that the intrusives are considered to be essentially coeval with the lavas. Although considered to be related, the Kalkrand basalts and associated intrusives in Namibia are comparatively isolated. In the subdivision of Eales *et al.* (1984) the Lebombo area is distinguished from the central area, and indeed it displays a very much more complex stratigraphy along its length.

The precise relation between these various igneous manifestations has long been uncertain, as has the duration of the magmatic activity that they represent.

The majority of older radiometric determinations on the Karoo rocks used the K/Ar technique, and the results displayed a considerable spread from 200 to 150 Ma. The similarity of pole positions obtained from all these rocks over this 50 my period contributed to the belief that a long interval of palaeomagnetic stillstand had occurred ('quasi-static interval' of Briden, 1967) for all of Africa.

Considerably more K–Ar data, and more recently, $^{40}Ar/^{39}Ar$ data, of the dolerites and basalts have been added since (Fitch & Miller, 1984), but these still show a comparable spread. Although fewer in number, Rb–Sr ages are generally somewhat more restricted, ranging from around 190 to 175 Ma. These include 180 ± 2 Ma for the Tandjiesberg sill in the Western Cape (Richardson, 1984) and 197 ± 11 Ma for the Birds River Complex in the northeastern Cape (Allsopp *et al.*, 1984a). In the Lebombo area, Rb–Sr dates on felsic rocks ranged from 175 ± 7 Ma for the Jozini rhyolite to 191 ± 9 Ma for Nuanetsi rhyolite in Zimbabwe (Allsopp *et al.*, 1984b). Ages for the Bumbeni and Kuleni Complexes, which intrude the Jozini rhyolites in Natal, are distinctly younger, between 140 and 145 Ma (Allsopp *et al.*, 1984b).

Recent dating in other continental flood basalt provinces (Duncan & Pyle, 1988; Renne & Basu, 1991; Renne *et al.*, 1992) have demonstrated the short-lived duration of the igneous activity, in marked contrast to the wide span of apparent ages reported in the above studies of the Karoo rocks. With

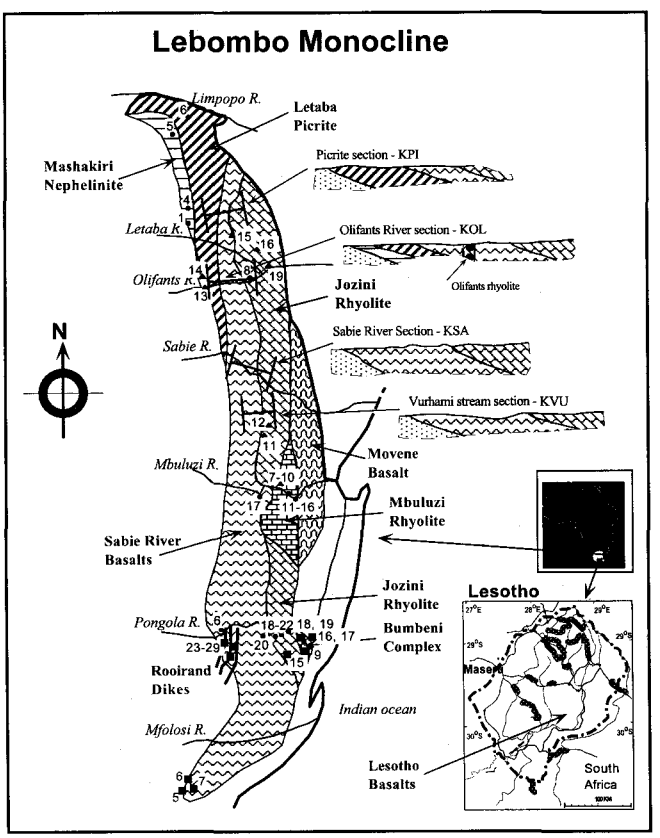


Figure 2 Sample locations in Lebombo. Numbered samples — Hargraves (black squares, 1987; triangles, 1988; dots, 1990); |—| symbols denote four stratigraphic sections sampled by Hooper's group. Inset: location of stratigraphic sections sampled in Lesotho.

the goal of clarifying this temporal dilemma, $^{40}Ar/^{39}Ar$ dating was undertaken on some of the samples collected in our palaeomagnetic study.

New dating

In the laboratory of T.C. Onstott at Princeton University, $^{40}Ar/^{39}Ar$ dating of samples collected in the Hargraves study employed laser-spot-fusion of plagioclase (or sanidine) grains in thick thin sections cut from palaeomagnetic cores. The least altered dyke samples with coarsest plagioclase grains were selected for irradiation based on petrographic examination of normal thin sections; but these still showed varying degrees of deuteric alteration. The results of these measurements are summarized in Table 2a. Excluding the Bumbeni syenite sanidines (two grains, incrementally heated), the dolerites still show an apparent spread in age from ~178 to 201 Ma, and the temporal uncertainty remains. The oldest age, however, clearly correlates with a high $^{40}Ar/^{36}Ar$. Apart from possible Ar leakage, the presence of excess argon clearly further complicates Ar dating of intrusives.

Duncan *et al.* (1997) have had more success with $^{40}Ar/^{39}Ar$ dating by incremental heating of lavas. Whole-rock chips and/or plagioclase extracted from 30 samples in the Karoo collection of Hooper *et al.* (1993) in Lesotho, Lebombo, and Namibia were analysed (plus three from Antarctica), and the results (using an age of 523.5 Ma for Mmhb-1, the flux monitor) are summarized in Table 2b. Although some scatter is observed, and the Jozini rhyolites are two to five Ma younger, the overwhelming majority of basalt samples from all three areas yield ages clustering around 183 ± 2 Ma. This is consistent with the recently published U–Pb (zircon and baddeleyite) ages of Encarnacion *et al.* (1996) for the New Amalfi dolerite sheet of 183.7 ± 0.6 Ma.

The best interpretation of all the available radiometric data is that, with the possibility of some earlier (~190 Ma) precursors, the vast bulk of the Karoo basalts were extruded or emplaced around 183 ± 2 Ma. The rhyolites are a few million years younger.

Palaeomagnetic sampling and measurement

In Hargraves' study, cores were drilled in the field, averaging six per site and oriented by sun and magnetic compass. The cores at each site were distributed over distances commensurate with the size of the outcrop, or the width of the unit being sampled. They were at least one metre apart, and usually five to ten metres (as in extensive road cuttings). These cores were cut into the usual 2.5 cm cylindrical specimens. Palaeomagnetic measurements were made in two laboratories: at Princeton University, and at the Geological Survey of South Africa in Pretoria.

Alternating field demagnetization was the method of choice for all samples. Stepwise demagnetization in 2.5 to 5.0 mT increments was continued up to 30 mT, and in 10 mT steps thereafter, or up to the point where serious Rotational Remanent Magnetism (RRM) problems negated further treatment. Thermal demagnetization was performed on individual specimens from most sites to confirm the carrier of the NRM, and on duplicate specimens from some sites where consistent vectorial results were not obtained. Data were analysed by linear regression (Kirschvink, 1980; see representative Zijderveld diagrams in Appendix) and averages calculated by Fisher and Bingham statistics (Onstott, 1980).

In the study of Hooper and co-workers, two to three cores were collected at most of the stratigraphically controlled sampling sites [six cores were collected at each of the sampling sites of the KAT section in northern Lesotho, with more detailed coverage in the polarity transition, and excursion zones (Rehacek, 1995)]. Following systematic AF demagnetization in ten steps to a maximum of 80 mT of representative specimens from each stratigraphic section sampled (see rep-

Table 3 Karoo paleomagnetic (south) poles

No	Designation	Symbol	Sites	Samples	Lat.	Long.	k	α
1	Lesotho lavas	LS	11	508N/91R	–74.4	92.8	104	4.5
2	Dolerite dykes	DD	22	110N/23R	–68.3	93.7	21	7.0
3	Hooper sills	TRA	14	23N/24R	–64.8	97.9	21	8.9
4	Mashikiri & Letaba	MAL	6(5+1)	12N/14R	–69.0	99.2	98	6.8
5	Sabie River basalts, Henthorn (1981)*	SR	23	46N/44R	–67.5	95.3	13	8.6
6	Jozini, Kruger	JK	2	9N	–70.2	94.7	331	13.8
7	Mbuluzi, Swaziland	MB	9	33N/4R	–45.6	98.6	27	10.2
8	Rooirand dykes	RR	8	26N/7R	–77.3	116.0	8	18.8
9	Jozini, Natal	JR	8	4N/40R	–70.7	111.2	29	10.5
10	Keetmanshoop sills	KEE	4	21N	–69.8	96.2	27	18.1
11	Hardap basalts (Kalkrand)	HAR	5	29N	–73.1	17.7	348	4.1
12	Bumbeni Complex	BU	4	14N/12R	–30.2	105.9	131	8.1
13	Cape Peninsula dolerites	CP	3	17N	–42.2	105.2	186	9.1
	MEAN of 1, 2, 3	KD	3		–69.2	95.2	267	7.6
	MEAN of 1, 2, 3, 4, 5, 9	KP	6		–69.2	98.3	416	3.3

*Henthorn's (1981) sampling of the Sabie River basalts along the Olifants River was more systematic and detailed than ours, and the pole calculated from his results (excluding transitional data) is included here. **These eleven 'sites' represent eleven distinct stratigraphic samplings each collected through the entire basalt column. (See Appendix 1)

resentative Zijderveld diagrams in Appendix), the balance were batch-demagnetized at a selected field (usually 20 or 30 mT). Sites where the α_{95} exceeded 20° were excluded from further consideration. All magnetic measurements were made in the Idaho Geological Survey Laboratory in Moscow, Idaho (Rehacek, 1995).

Structural corrections

The only corrections applied to the palaeomagnetic data are for sites affected by the east-dipping Lebombo monocline in Natal, Swaziland, and the Kruger National Park. For the most part, stratification was rarely evident in the outcrops sampled, and information as to structural attitude was obtained from geological maps, and by personal communication from John Bristow who has had considerable experience throughout that area.

The structural corrections in Natal and Kruger Park were not large (5° to 20°, easterly), but in the Mbuluzi Gorge in Swaziland, flow banding at some sites suggested dips up to 90°. The *in situ* site vectors were scattered in this area, and clearly considerable structural correction was required. A variety of rotations was tried for each site in Swaziland, and those most consistent with available field data and which yielded what were judged to be the most consistent data, were selected. The actual rotations applied are indicated in the Appendix tables, together with the 'before' and 'after' site vectors.

Palaeomagnetic results

The detailed results are presented in Appendices 1 and 2, and summarized in Table 3. In the Lesotho study, in some cases successive flows were found to have almost identical vectors, implying that they were erupted over an interval too short to record significant secular change. To avoid unduly biasing the overall averages, the method of McFadden & Lowes (1981) was employed to establish whether or not the directions of successive flows were statistically indistinguishable. If a series was found to be so, the indistinguishable flows were grouped into a single 'cooling unit'. In all cases, individual palaeomagnetic cooling units were restricted to individual geochemical units. Further, in the six stratigraphic sections, which included a reversal, the reversal test (McFadden & Lowes, 1981; McFadden & McElhinny, 1990) shows that the antipode of the mean reversed polarity direction is not distinguishable from the mean of the normal polarity direction at the 95% confidence level (see table 5.2 in Rehacek, 1995).

Two Lesotho poles were calculated: (a) a mean palaeomagnetic pole for each of the 13 basalt sections sampled (see Table 3); 11 of these were then averaged to give a Lesotho lava pole; and (b) a mean pole for all sites in each of the geochemically distinct stratigraphic units — some extending over 20 000 km² (Figure 3) as defined in Rehacek (1995). An average of these also yielded a Lesotho lava pole which is indistinguishable from the first. The mean section pole (LS) is shown in Table 3 and Figure 3.

In Hargraves' study, Karoo dolerite dykes and sills at 22 sites throughout the Central Karoo basin and into Natal (Figure 1) were averaged to yield a 'Dolerite dyke' pole (DD, Table 3). Fourteen of 49 sites collected by Hooper and co-workers in dykes and sills in the Free State, Mpumalanga, and

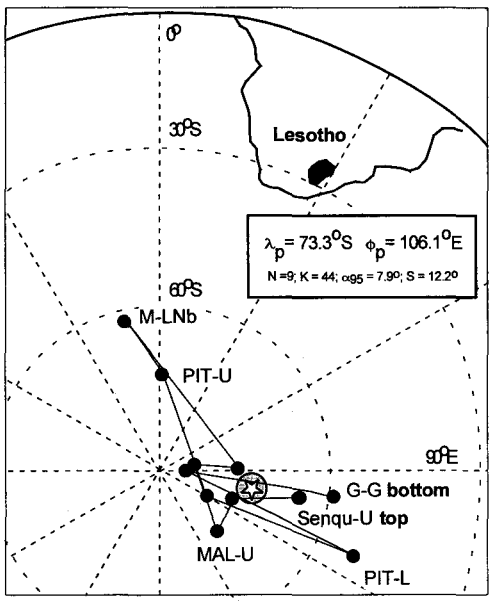


Figure 3 Pole positions for individual localities investigated in this study (abbreviations as in Table 3).

Gauteng (stars in Figure 1) had α_{95} less than 20°, and a mean for these was calculated (TRA, Table 3).

Whereas the mean pole from Keetmanshoop sills in Namibia (KEE) was similar to that of the Lesotho lavas and the dolerite dykes (see Table 3), that for the Kalkrand basalts (MAR) is significantly different. A small cryptic tectonic rotation of ~15° to the east is suspected as the cause of this discrepancy, but in the absence of definitive evidence, this result has not been included in the final averages.

In the Kruger Park, results from the sampling of both Hooper and co-workers and Hargraves were rather poor (see Appendix). Hooper's group collected samples (2 cores per site) through the entire stratigraphic column along four river sections (see Figure 2), but of 55 sites, only 28 gave useful results (see Appendix). Although important for magnetic stratigraphy, most of these data were not considered acceptable for VGP determinations. Hargraves' sampling concentrated on the Jozini rhyolites (seven sites) and Mashikiri nephelinites (five sites), plus one in the Letaba picrite — four to seven cores per site were collected (see Figure 2). Only two (both Normal) of the seven sites in the rhyolites were useful (the two sites, 88-16 and 88-19, which yielded intermediate directions, have been omitted), and their mean (see Table 3) is similar to that of the Jozini rhyolites in Natal; this result, however, has not been used in any overall averages. Consistent results were obtained from four of the five Mashikiri nephelinite sites, but at one of these (90-4) the direction appeared intermediate and was discounted. At another (90-1), however, two distinct components (N and R) could be isolated (see Appendix), and these, plus the remaining sites and one in the picrite (90-6, R), have been combined with the KOL Site '18 of Hooper and co-workers (see Appendix) to give a pole MA (see Table 3).

Hargraves' results from the Jozini rhyolites in Natal (JR) and the Mbuluzi rhyolites in Swaziland (MB) — both sampled in fresh road or railroad cuts — gave fairly straightforward results. The means were calculated for these (Table 3).

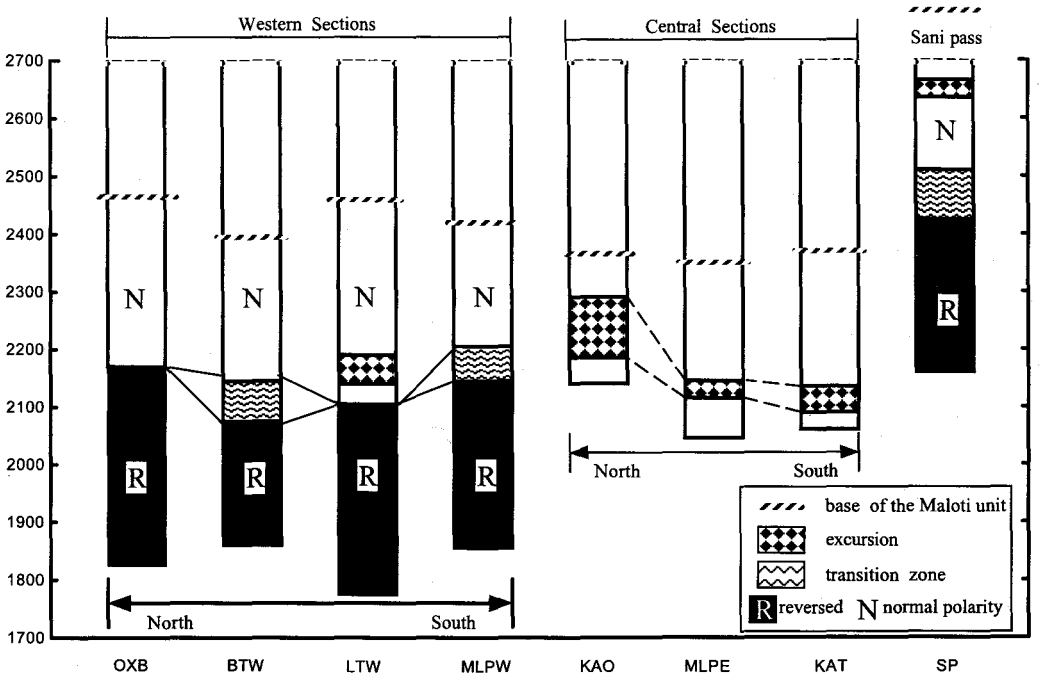


Figure 4 Magnetostratigraphy of individual sections in Lesotho basalts. Note that the base of the normally oriented flows is lower in Central Sections than in Western Sections and in Sani Pass section on the eastern edge of the Lesotho basalt cover.

Results from the eight sites (fresh road cuts) in the Rooirand dyke swarm in Natal showed considerable scatter. Polarities are mixed and some sites yielded intermediate directions. The sheeted nature of this swarm, with dykes cutting and reheating older dykes, doubtless contributes to the complexity, but the mean pole calculated (RR) is not significantly different from the main Lesotho lavas and dolerite dykes. This pole was not used in the overall average.

The Bumbeni Complex in Natal (four sites) and the Peninsula dolerites in the Cape (two sites) are known to be significantly younger than the main Karoo magmatism. In the Bumbeni Complex, both syenite and perlite (from Nxwala mine) were collected, and the presence of mixed polarities suggests that a significant time interval is represented.

The VGP directions from the two Peninsula dolerite dykes were clearly distinct from the bulk of the Karoo results, but these data are too few for a palaeomagnetic pole.

Magnetic stratigraphy

The palaeomagnetic reversal first described by Van Zijl *et al.* (1962) in the lower part of the basalt sequence in Lesotho has now been traced across the whole Lesotho Kingdom and is present in all sections sampled down to the lower contact with the Clarens Sandstone (Figure 4). The reversal is characterized by a brief transition zone of one to three flows, between which there is no evidence of erosion. This latter evidence suggests a rapid rate of basalt extrusion. Certainly, the presence of only one reversal in the whole 1400-meter-thick basalt pile in Lesotho during a period of frequent reversals (Gradstein *et al.*, 1994) requires an unusually rapid rate of extrusion. This is in accord with Duncan *et al.*'s (1997) $^{40}Ar/^{39}Ar$ ages (183 ± 2 Ma) (Table 2b) that do not differ significantly from top to bottom of the Lesotho basalts.

A short palaeomagnetic excursion consisting of a brief switch to reversed polarity (inc. < 45°) is also evident immediately above the main reversal in Lesotho (Figure 5) and can be detected in the data of Van Zijl *et al.* (1962). Its high intensity and short duration are typical of such excursions (Merrill & McFadden, 1990). The excursion is confined to a strati-

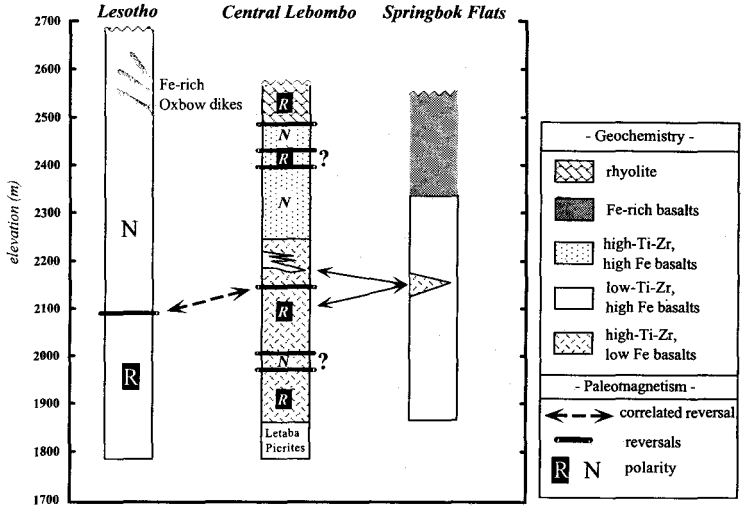


Figure 5 Geochemical and paleomagnetic correlations between several areas of Karoo Volcanic Province (simplified after Marsh *et al.*, in press). All basalts in Lesotho and Springbok Flats (one exception) are low-Ti Broken line points to the reversal in Lesotho and Lebombo correlated by Henthorn (1981). Reversals marked with the question mark (?) are recorded in a single flow only. The scale on the left (elevation in metres) relates to Lesotho only.

Table 4 Compilation of magnetic stratigraphy in Central Lebombo region

Lebombo Monocline magnetic stratigraphy								
	Kruger						Swaziland	Natal
	Henthom (1982)	KPI	KOL	KSA	KVU	Hargraves	Hargraves	Hargraves
Bumbeni Complex								N/R
Younger Dyke							R	
Mbuluzi Rhyolite							N	
							(R)	
Jozini Rhyolite	N				N		N	N
			R	R		R		R
				N	N	N		Rooirand N/R
	(R)		R	R				R
	N		N	N	N			
Sabie River Basalt	R		R	R	R			
				(N)	(N)	R		
		R	R	R				
Letaba Picrite	R	R				R		
Mashikiri Nephelinite			N			N		

Double line in Sabie River Basalt section marks the reversal (R – N) correlated with the reversal in Lesotho basalts. The broken line marks possible excursion in these basalts

graphic unit defined by its specific composition (low Zr/Nb, known as the Mafika–Lisiu unit; Rehacek, 1995) and provides another marker horizon for stratigraphic control in the Lesotho basalt sequence.

The polarity of all samples from central and southern Namibia (Keetmanshoop dolerites and a section through the Hardap section of the Kalkrand basalts) have normal polarity. An apparent excursion represented by a single flow at Hardap could correlate with the excursion above the main reversal in Lesotho, which has a similar age (183 ± 9 Ma) and similar pole position. A direct correlation is supported by the geochemical similarity in that the basalts in both cases are the low-Ti variety (Marsh *et al.*, in press).

The dykes and sills sampled in the central and southern part of the Karoo basin are all normally polarized, with reversals appearing only toward the east, in KwaZulu–Natal and Gauteng, between Lesotho and Swaziland. The temporal significance of the change is not clear.

The greatest compositional diversity within the Karoo Igneous Province occurs along the Lebombo monocline (Figure 2). In the north the Mashikiri nephelinites lie at the base, with the Letaba picrites, the Sabie River basalts, and then the Jozini rhyolites lying above. In Swaziland, the Mbuluzi rhyolites overlie the Jozini rhyolites (Cleverly & Bristow, 1979). Unfortunately, the Movene basalts in Mozambique, which overlie these rhyolites, were not sampled due to the civil war in that country at the time of sampling.

Possibly as many as seven distinct reversals of the magnetic field are recorded in the composite Lebombo sequence (Table 4). The greatest uncertainty occurs in Kruger Park where outcrops are natural in contrast to Swaziland and Natal where deep road or railroad cuts were sampled. The magnetostratigraphic correlation between the various sections (Table 4) is tentative. A change from normal to reversed is apparent in the Mashikiri–Letaba succession, the reverse sequence evidently persisting through the Letaba picrite and into the lower part of the Sabie River basalts. At least three reverse to normal changes occur within the Sabie River basalt, but all three are present in only one section (KSA) and only one reversal is present in all three sections. Henthorn (1981) found a similar situation in his single section and suggested that some of the reversals might be due to intrusions, rather than flows, being sampled in this poorly exposed area, but this is probably not so for the current study. The main change from reverse to nor-

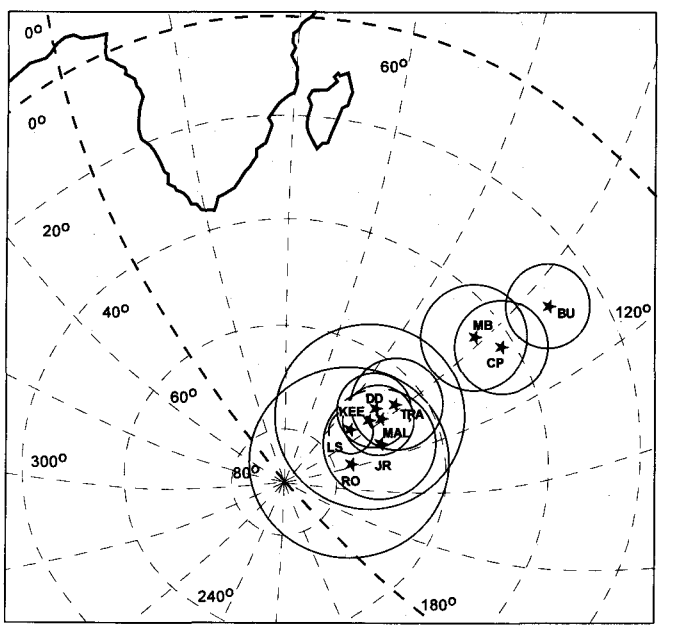


Figure 6 VGPs calculated from individual stratigraphic units in Lesotho. Note relatively small secular variation within the 1400-m-thick basalt sequence. Outlying M-LNb and PIT-U contain the reversal while the uppermost stratigraphic unit M-U was probably affected by IRM overprint caused by lightning. These units were excluded from paleopole calculations.

Table 5 Early Jurassic paleomagnetic poles from Antarctica and West Africa

	Abbr.	No. of Sites (samples)	N. Pole Lat.°N	Long.°E	(k)	α_{95}	Probable magnetization age
Ferrar dolerites (1)			54.3	40.9	dp=4.5	dm=5.0	180±5
Rotated to Africa (2)	AN		69.9	257.3			
Foam Zguid dyke, Morocco (3)	FZ	5	58.0	259.0	(316)	4.0	182–187, K–Ar
Draa Valley sills, Morocco (3)	DV	16	65.5	230.5	(70)	3.5	180–186, K–Ar
North Mauretania dolerites (4)	NM	13	69.4	232	(99)	(4.1)	187±5, K–Ar
South Mauretania dolerites (4)	SM	13	71.4	240.2	(47)	(6.1)	187±5, K–Ar
Liberian dykes (5)	LD	25	68.5	242.4	(31)	(5.3)	173–192, K–Ar
Freetown Complex, S.L. (6)	FR	10*	80.9	224.6	(43)	(4.8)	193±3, Rb–Sr
La Reculée sediments (7)	LR	8	70.9	235.2	478	(2.3)	180–210 Ma (strat.)

(1) Mcintosh *et al.*, 1986; (2) Martin & Hartnady, 1986; (3) Hailwood & Mitchell,1971; (4) Sichler *et al.*, 1980; (5) Dalrymple *et al.*, 1975; (6) Briden *et al.*, 1971; Beckinsale *et al.*, 1977; (7) Kies *et al.*, 1995

Brackets around k and/or a95 when data are available only for mean of directions, not mean of poles

mal recorded by Henthorn (1981) toward the base of the Sabie River basalts along the Olifants river was the basis of his tentative correlation with the equivalent reversal in the Lesotho basalts, and this is equated with the reversal found in all three of our sections. This lower reversal occurs below the low-Ti basalt horizon in the Sabie River unit while the upper (excursion?) occurs in the high-Ti–Zr–Fe basalts. The whole Lesotho sequence is of low-Ti type, so this is much the most obvious correlation to make (see Figure 6).

The bulk of the Jozini rhyolites in Kruger Park and northern Natal are reversed (as is the Sabie River basalt immediately underlying the rhyolites in Natal). Bristow (1982) distinguishes seven ignimbritic flow units in the Jozini rhyolite sequence in northern Natal. The lower five of these are reversed, the sixth is transitional, and the upper is normal. The bulk of the Mbuluzi rhyolites is normal [one rhyolite site and a younger (= Movene?) basaltic dyke are reversed].

Dykes of the Rooirand dyke swarm in Natal (Armstrong *et al.*, 1985) are of mixed polarities. They cut the Sabie River basalts, and appear to preceed the overlying Jozini rhyolites.

The magnetic polarity changes through the sequence along the Lebombo monocline is not entirely clear, due in large part to poor exposures. But at least one significant conclusion is apparent. The compositional diversity, the minimum of four and possibly as many as seven magnetic reversals, and the somewhat greater range of $^{40}Ar/^{39}Ar$ ages (> 184 Ma to 178 Ma) all imply that this sequence of flows erupted over a longer period of time than did the basalts in Lesotho (183 ± 2 Ma).

The magnetic stratigraphy along the Lebombo monocline is best shown in Table 4 and its relation to that in Lesotho in Figure 6. Given the frequency of field reversals over this period (175 – 190 Ma), averaging about 1.5 reversals per million years (Gradstein *et al.*, 1994), the presence of so many reversals is inevitable. What is surprising is the predominance of normal polarities in the dykes and sills of the central area of the Karoo and the upper part of the Lesotho basalts. The simplest interpretation is that most of the Lesotho lavas and the majority of dolerites from the Karoo Basin, in addition to the flows and dolerites from southern Namibia, were erupted within a single normal polarity epoch, probably within one million years, about 183 Ma. In the northern part of the Lebombo monocline the eruptions started somewhat earlier and continued for a longer time.

Palaeomagnetic pole positions

Palaeomagnetic results are summarized in Table 3 and Figure 3, with details in Appendices 1 and 2.

To test whether secular variation has been adequately averaged in the poles calculated, the angular dispersion (S) of each VGP dataset was compared to the compilation of palaeosecular variation presented by McFadden *et al.* (1991). The dispersion expected for the 110 – 195 Ma age interval, and for latitude ~300, is 18 ± 3°. Angular dispersion (S) of poles calculated from a larger dataset (Lesotho, Sabie, Dolerite; Figure 3) is well within the predicted interval for the given latitudes. The angular dispersion of the poles calculated from datasets containing only a few VGPs (Mashikiri–Letaba, Jozini, HAR, KEE) are too low to satisfy this criterion (Appendix 1) and should be judged accordingly.

With the exception of the poles from the Cape Peninsula dykes, from Bumbeni Complex, and from the Mbuluzi rhyolites in Swaziland, the poles for all units sampled in this combined study, from the earliest nephelinites in the northern part of the Lebombo monocline, through the huge volumes of tholeiitic basalts and dolerites in the central Karoo as far southwest as Namibia (the Hardap lavas have been excluded), to the Jozini rhyolites, are very similar (Table 3, Figure 3). The Cape Peninsula dolerites (~132 Ma, K/Ar, Reid *et al.*, 1991) and Bumbeni Complex (145 – 147 Ma, Allsopp *et al.*, 1984b; this study) are known to be considerably younger than 180 Ma and their poles are significantly different. The Mbuluzi rhyolite (Bristow & Cleverly, 1983), although stratigraphically younger than the Jozini, was thought to be penecontemporaneous and was not expected to yield a significantly different pole. In fact, the pole is significantly removed in the direction of the younger poles (Figure 3), and so either the Mbuluzi is significantly younger than the bulk of the Karoo basalt province, or its eruption coincides with the onset of more rapid drift. Further palaeomagnetic study, and dating of the Mbuluzi rhyolites and the Movene basalts, is needed.

In conjunction with the previously published data, the new palaeomagnetic and $^{40}Ar/^{39}Ar$ results are consistent with the hypothesis that the entire Karoo Igneous Province through the Jozini rhyolites was emplaced between about 190 Ma and 175 Ma, with the great majority of flows erupted in a comparatively brief interval of about 183 Ma. The older age data indicating a spectrum from 150 to 200 Ma appears incorrect. If near coincidence of pole positions is used as a basis for selection, then a mean pole for this time using both old and our new data can be calculated (Table 3). Six independent pole determinations, each the average of numerous sites, in different units, yielded an average south pole at latitude = –69.2, longitude = 98.3, α = 3.3, to which an age of 183 ± 5 Ma was assigned.

Discussion

West African palaeomagnetic poles of similar vintage to the Karoo poles (Table 5) are compared in Figure 7. These appear to be consistently displaced to the west of the Karoo poles (Hargraves, 1988; Kosterov & Perrin, 1996) and a detailed analysis of this contrast and its tectonic significance is in preparation.

Stratigraphically older Mesozoic sedimentary rocks from southern Africa have been studied palaeomagnetically but, as Ballard *et al.* (1986) have shown, remagnetization, perhaps related to the Karoo igneous activity, is pervasive.

The only other Jurassic pole which emerges from this and previous studies is represented by the Bumbeni Complex (145 – 147 Ma). It is very similar to the Swartruggens kimberlite-fissure pole reported by Hargraves (1989), which is of the same age (~145 Ma, Allsopp *et al.*, 1984b), and also includes both polarities. Their means are almost indistinguishable, and so a mean pole from the two studies (N = 6 sites) has been calculated at latitude = –31.7, longitude = 104.3, k = 94, α = 6.3. This, the authors submit, is the best determined upper Jurassic (145 Ma) pole from southern Africa. The age of the Mbuluzi rhyolite is not known. Its pole is located roughly between the Karoo and Bumbeni–Swartruggens poles, perhaps suggestive of an intermediate age. The Cape dolerites have been dated at 132 ± 6 Ma (Reid *et al.*, 1991), but the pole reported here, represented by only two, or at most three, sites, is at best a VGP.

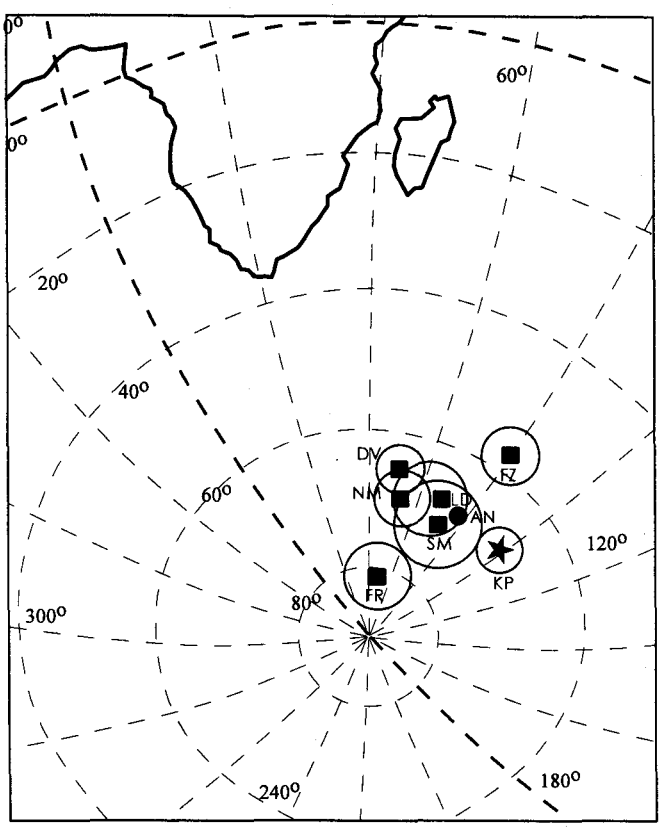


Figure 7 Comparison of the Karoo pole (star — this study) with West African poles (squares) and Antarctic pole (circle). See Table 5.

The most pertinent intercontinental comparison to make with our Karoo pole is with the Ferrar dolerites in Antarctica. Encarnacion *et al.* (1996) have reported almost identical and high precision U–Pb (zircon, baddeleyite) ages of ~184 Ma for two Ferrar dolerites and for the New Amalfi dolerite sheet in the southeastern Karoo province. A correlation with the basaltic eruptions in the Karoo is further supported by a new Ar–Ar age for the Kirwan basalts in Antarctica (183.7 ± 0.6 Ma; Duncan *et al.*, 1997; Table 2b) which, as Harris *et al.* (1990) have shown, are geochemically identical to the low-Ti basalts of Lebombo, consistent with their close proximity before Gondwana breakup. Within experimental limits these ages are the same as those reported here for the main eruptions of Karoo basalts, and a comparison of poles from the two areas is appropriate. Models for the Gondwana positioning of Antarctic with respect to Africa vary somewhat (Smith & Hallam, 1970; Norton & Sclater, 1979; Martin & Hartnady, 1986). That of Martin & Hartnady (1986) is the most recent, and was derived specifically for the Africa–Antarctica reconstruction. The mean Antarctica pole (seven studies, 141 sites, > 133 Normal; Pole = 54.3S, 220.9E, dp = 4.5, dm = 5.0; McIntosh *et al.*, 1986) rotated according to the Martin & Hartnady model is at latitude = –69.9, longitude = 77.3, dp = 4.5, dm = 5.0 (Table 5, Figure 7). The difference between the Antarctic and Karoo poles is within error, but clearly the Antarctic pole seems closer to the West African poles than to the Karoo (Figure 7). Whether the rotation pole used for the Antarctic pole is incorrect, or the apparent difference between the West African and southern African poles alluded to above is not significant, remains unclear.

Conclusions

This study supports the contention that the bulk of the Karoo Igneous Province was emplaced very rapidly, largely during one single polarity epoch, about 183 Ma ago. The volcanism was virtually contemporaneous over most of southern Africa and can be correlated with similar volcanic activity in Antarctica.

Two palaeomagnetic poles were determined for the Karoo volcanic rocks from the extensive database now available. A particularly robust palaeopole position at latitude = –69.2, longitude = 98.3, k = 416, α = 3.3 for Karoo Volcanic Province at 183 ± 5 Ma has been calculated. It is noteworthy that this pole is not significantly different from those obtained in earlier palaeomagnetic studies of Karoo igneous rocks (e.g. Van Zijl *et al.*, 1962; see Table 1). A relatively robust late Jurassic (~145 Ma) pole at latitude = 31.7, longitude = 104.3, k = 94, α = 6.3 is calculated by combining bipolar results

from the Bumbeni Complex (syenite and Nxwala perlite) and the Swartruggens kimberlite fissures.

The magnetostratigraphic record from the three volcanic sequences investigated (Kruger Park, Lesotho, and Namibia) permits the correlation of the eruptive history across the southern part of the Karoo province. The polarity sequence of the Sabie River basalt and Lesotho basalt sequences probably record the same palaeomagnetic reversal (reverse to normal) as suggested by Henthorn (1981). In addition, the palaeomagnetic record from dolerite sills and dykes sampled between Lesotho and central Lebombo show similar reversed and normal polarities. The normal orientation of basalts and dolerites in the southern Karoo basin and central Namibia can be correlated with the upper parts of the basaltic sequence in Lesotho. From the data presented, it is concluded that the main part of the basaltic volcanism of the Sabie River basalt, the Lesotho basalt, and the basaltic sequence of the Hardap section in the Kalkrand basalts, were erupted over the same short time period, which included a single palaeomagnetic reversal.

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Appendix 1

Results of the Hooper *et al.* (1993) study

		Site							Pole	VGP
Site	Polarity	Lat.	Long.	N/n	D	I	k	α95	Lat.	Long.
Lesotho Section pole				No stuctural correction						
OXB		–28.8	28.6	121/52	160.4	57.8	23.9	8.7	–71.1	82.9
BTW		–28.8	28.4	67/56	155.9	53.0	26.8	9.5	–68.9	99.4
LTW		–28.9	28.3	74/52	160.5	54.0	37.3	8.0	–72.6	94.6
MLPW		–29.0	28.4	145/127	154.3	54.7	56.9	4.2	–67.4	95.8
MKH		–29.1	28.9	69/52	157.9	56.2	80.7	5.8	–70.0	90.7
LTE		–29.1	28.5	68/49	168.7	55.2	29.0	9.7	–78.4	80.5
MLPE		–29.1	28.5	75/69	156.5	53.7	37.2	7.6	–69.4	98.0
KAT		–29.2	28.5	193/148	186.9	47.5	8.7	18.5	–83.9	–65.3
SOM		–29.8	27.1	91/56	163.5	53.9	16.8	8.5	–75.3	94.5
SP		–29.6	29.3	80/68	162.4	53.6	35.6	7.4	–74.4	97.6
ON		–28.3	30.3	57/37	163.3	52.8	57.5	5.1	–74.8	96.6
Lesotho Unit pole				No stuctural correction						
G-G	R	–29.0	28.5	6/4	144.4	54.7	148.3	6.2	–59.3	97.5
LET	R	–29.0	28.5	6/4	175.6	50.3	158.5	6.0	–85.7	88.6
PIT-L	R	–29.0	28.5	8/4	137.3	43.6	127.9	6.7	–52.1	113.4
PIT-M	R	–29.0	28.5	6/4	169.9	47.6	31.5	13.5	–81.1	118.0
PIT-U	T	–29.0	28.5	38/23	191.2	62.6	3.7	18.2	–72.6	0.8
M-LNb	T	–29.0	28.5	53/43	206.5	64.9	4.0	12.6	–62.7	–13.2

Appendix 1 (Continued)

		Site							Pole	VGP
Site	Polarity	Lat.	Long.	N/n	D	I	k	α95	Lat.	Long.
M-L	N	-29.0	28.5	145/115	164.9	54.4	8.6	4.8	-75.9	89.7
MAL	N	-29.0	28.5	86/64	174.1	52.3	7.4	7.0	-83.7	79.4
MAL-U	N	-29.0	28.5	30/22	165.0	41.1	8.6	11.2	-75.5	137.0
SENQU	N	-29.0	28.5	278/198	164.7	49.0	7.2	4.0	-76.6	111.0
SENQU-U	N	-29.0	28.5	176/108	151.2	53.0	15.2	3.6	-65.1	100.7
Dolerite pole		No stuctural correction								
TRA-02	R	-27.0	28.7	4	173.3	49.2	120.3	9.1	-83.3	89.3
TRA-24	N	-28.3	28.7	2	111.6	74.3	36.7	12.4	-34.9	62.4
TRA-27	N	-28.3	28.7	4	143.1	55.2	39.2	14.9	-58.0	95.8
TR-32	R	-27.8	29.5	3	140.8	47.0	422.7	6.0	-55.6	109.4
TRA-37	R	-28.6	29.8	4	170.7	56.8	47.0	13.6	-78.2	68.8
TRA-57	N	-27.5	31.7	4	152.1	45.1	698.4	3.5	-65.2	117.2
TRA-63	N	-27.3	31.5	4	170.3	56.4	41.1	19.5	-77.3	69.4
TRA-72	N	-27.3	31.5	2	156.0	47.1	65.1	11.5	-68.8	113.4
TRA-79	N	-27.3	30.4	3	154.7	54.3	808.9	3.2	-67.1	94.8
TRA-82	R	-27.2	29.9	3	140.4	32.5	31.5	16.6	-52.3	126.1
TRA-86	N	-26.9	29.3	4	159.8	62.4	11.3	8.8	-66.5	68.3
TRA-92	R	-27.3	29.5	5	133.5	35.2	19.9	17.6	-46.8	119.7
TRA-93	R	-27.2	29.7	3	141.2	30.3	1789.0	5.9	-52.5	128.4
TRA-94	R	-26.9	29.2	2	165.2	64.8	1309.5	6.9	-67.0	55.8
Sabie River basalt		Corrected for bedding tilt of 5° E								
Sabie-OL5	R	-24.0	31.8	2	163.8	40.3	182.0	18.6	-75.1	122.5
Sabie-OL4	N	-24.0	31.8	2	223.0	27.0	2241.0	5.3	-48.4	-63.7
Sabie-OL3	N	-24.0	31.8	5	148.3	52.4	17.0	18.9	-60.9	96.5
Sabie-OL2	R	-24.0	31.8	2	186.4	73.1	416.0	12.3	-55.0	26.0
Sabie-OL1	R	-24.0	31.8	2	117.9	53.4	593.0	10.3	-35.6	96.1
Sabie-SA5	N	-25.1	31.9	4	197.2	46.5	36.0	15.5	-74.4	-44.4
Sabie-SA4	N	-25.1	31.9	5	174.1	59.7	206.0	5.3	-73.8	48.1
Sabie-SA3	R	-25.1	31.9	4	203.5	46.8	28.0	17.6	-68.8	-45.0
Sabie-SA1	R	-25.1	31.9	2	173.1	53.3	1093.0	7.6	-79.4	64.6
Sabie-VU4	N	-23.8	31.6	3	187.6	26.1	1178.0	3.6	-77.7	-111.5
Sabie-VU3	N	-23.8	31.6	2	150.3	57.4	331.0	13.8	-61.0	85.3
Sabie-VU2	R	-23.8	31.6	4	151.1	66.2	28.0	17.8	-56.4	66.9
Sabie-VU1	N	-23.8	31.6	2	151.4	59.0	175.0	19.0	-61.1	81.2

OL, SA, and VU designate geographic location of the sites; Olifants river, Sabie river, and Vurhami creek sections, respectively (Figure 2). The sample numbers in each section increase with stratigraphic height.

Mashikiri nephelinites & Letaba picrites					Corrected for bedding tilt of 5° E					
Mashikiri	N	-24.0	31.8	2	161.9	47.0	257.1	15.6	-73.2	103.4
Letaba	R	-23.8	31.6	3	169.9	51.0	45.0	18.6	-78.1	77.9
Kalkrand basalts		No stuctural correction								
HAR01	N	-24.5	17.8	6/4	182.1	58.6	750.6	9.1	-75.1	11.6
HAR03	N	-24.5	17.8	6/6	179.3	62.5	3615.8	1.3	-70.7	19.4
HAR04	N	-24.5	17.8	6/5	180.0	63.1	2004.5	5.6	-69.9	17.7
HAR06	N	-24.5	17.8	9/6	184.9	60.9	978.8	3.9	-72.1	5.8
HAR11	N	-24.5	17.8	9/8	174.2	56.5	778.4	4.4	-76.5	37.9
Keetmanshoop sills		No stuctural correction								
KEE05	N	-26.5	18.5	6/4	137.6	43.9	1290.9	7.0	-52.1	100.0
KEE06	N	-26.5	18.5	6/5	159.7	48.9	870.5	8.5	-71.8	93.2
KEE09	N	-26.5	18.5	12/8	152.5	50.3	36.9	15.3	-65.5	91.2
KEE12	N	-26.5	18.5	6/4	179.5	44.2	631.2	10.0	-89.2	159.9

Appendix 1 (Continued)

Pole positions and associated statistics for individual localities in the study of Hooper and co-workers

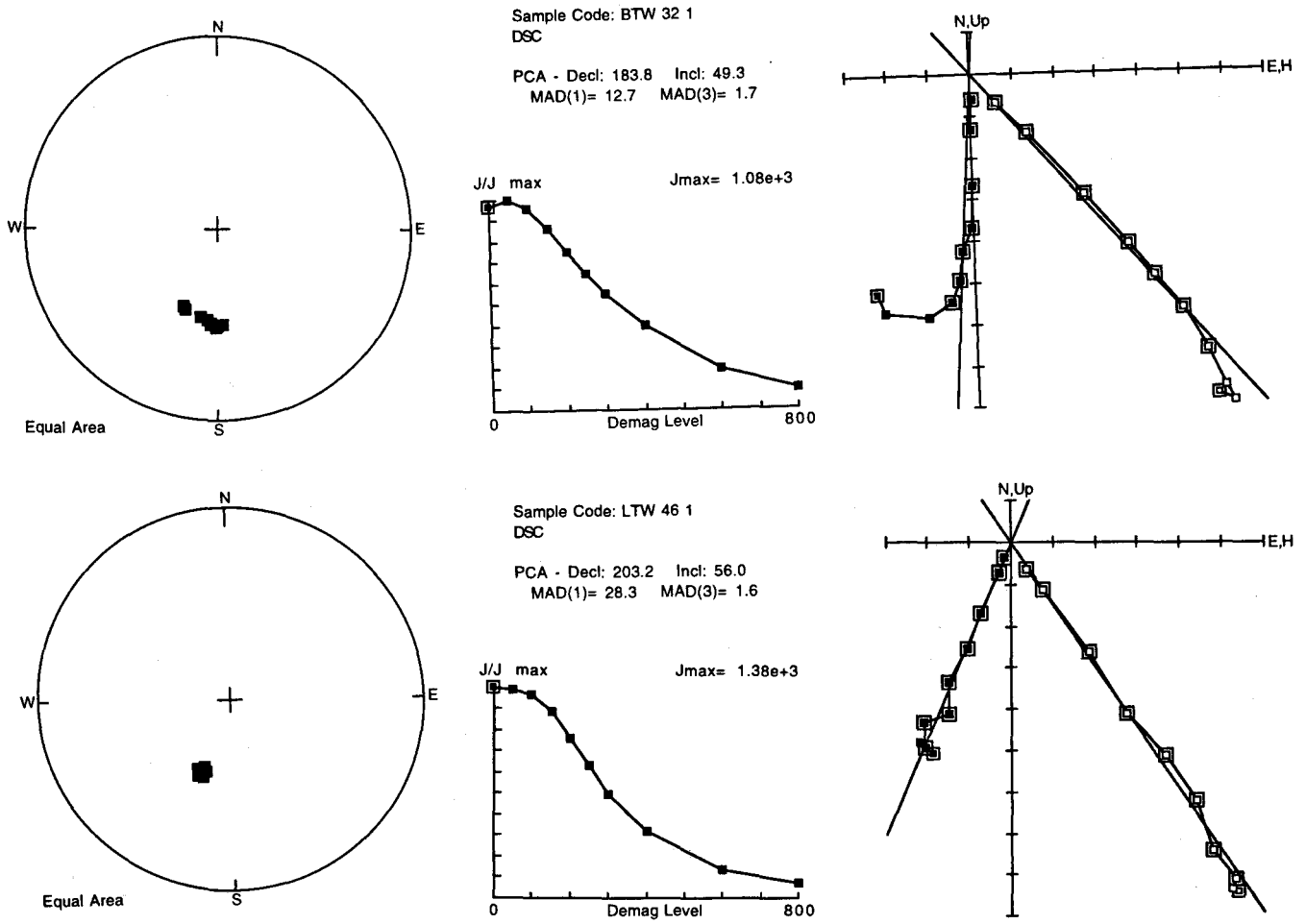
Locality	Long.	Lat.	N	R	K	S	a95
Lesotho Section	92.8	–74.4	11	10.9	104	8.0	4.5
Lesotho Unit	106.1	–73.3	9	8.8	44	12.2	7.9
Dolerite TRA	97.9	–64.8	14	13.4	21	17.6	8.9
Sabie River basalt	57.7	–76.2	13	11.7	9	26.4	14.3
HAR	17.7	–73.1	5	5.0	348	4.3	4.1
KEE	96.2	–69.8	4	3.9	27	15.6	18.1

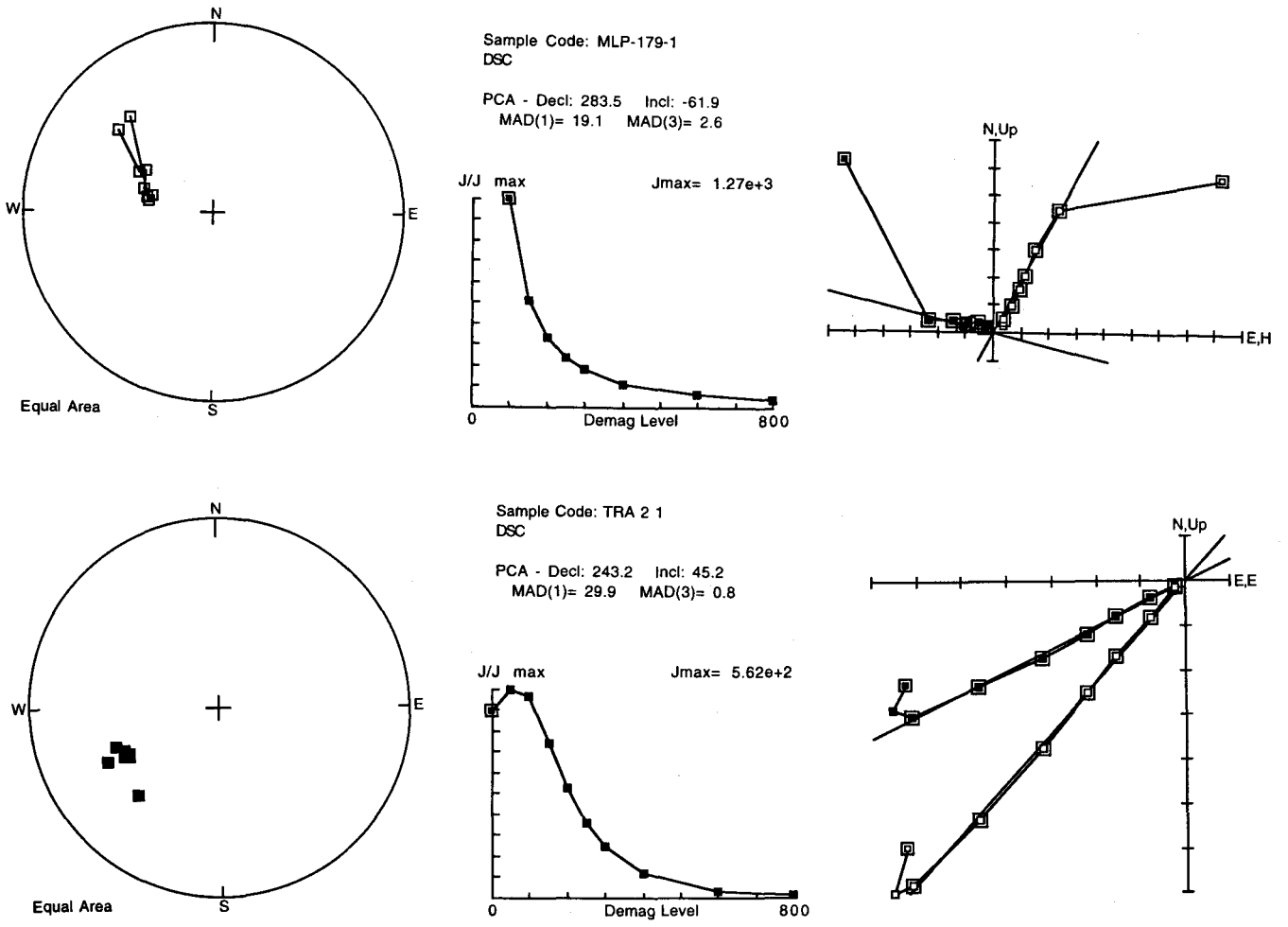
Appendix 1 (Continued)

Representative demagnetization results:

From left to right: Equal-area stereographic projection, normalized intensity versus AF; Zijderveld plot; PCA vector with MAD values shown beneath sample ID in upper centre

Core-sample			
	BTW 32-1	Lesotho lava	R
	LIW 46-1	Lesotho lava	R
	MLP 179-1	Lesotholava	N
	TRA 2-1	Dolerite sill, Gauteng	R





Appendix 2 Karoo dolerite dykes (D) and sills (S)

		Site						Bedding	Structurally corrected		Pole		
Site	No.	Lat./Long.	N	D	I	k	α	dip/azimuth	D	I	Lat.	Long.	α
D	85-1	–28.8, 24.7	6/6	358.7	–49.9	300	3.9				82.3	267.5	4.6
S	12	–30.3, 23.2	6/4	341.3	–56.5	173	7.0				72.8	264.5	9.6
D?	13	–30.8, 24.1	6/4	329.7	–54.4	126	8.2				64.4	276.2	10.3
D	14	–31.4, 24.1	6/5	329.3	–53.6	45	11.5				64.4	288.9	13.4
D	15	–31.9, 23.1	6/6	34.1	–63.5	19	15.8				61.2	149.8	24.1
S	16	32.3, 22.5	6/6	356.5	–57.9	104	6.6				83.0	225.8	9.3
D	17	–31.7, 21.7	5/5	319.1	–61.7	179	5.7				55.9	260.5	8.4
S	18	–32.2, 21.1	6/6	330.0	–48.9	107	6.5				64.1	288.1	8.1
S	19	–32.4, 20.6	6/6	331.7	–50.0	400	3.4				65.9	286.6	3.9
S	22	–32.6, 26.9	6/6	321.2	–62.4		7.1,13.3*				57.9	264.4	8.0
			6/6	305.2	–33.6	84	7.3						
S	23	–32.9, 27.5	6/6	331.0	–56.2	480	3.1				65.8	278.7	3.5
D	24	–32.4, 28.7	6/5	328.3	–68.0	125	6.9				59.9	250.0	10.8
D	25	–32.4, 28.7	6/6	326.6	–53.6	58	8.9				62.0	284.0	8.6
D	26	–32.2, 27.1	6/6	No significant grouping									
D	27	–32.8, 28.0	6/6	301.6	–50.1	13.2	18.0						
S	29	–31.9, 25.4	6/5	334.3	–69.2	253	4.8				62.1	239.3	8.0
S	30	–31.5, 24.0	6/4	101.3	–66.9	7	38.5				18.9	166.9	57.8
S	31	–30.7, 25.1	6/5	352.4	–49.1	25	15.7				84.0	293.7	18.4
D	32	–30.0, 25.3	6/6	346.8	–56.5	89	7.1				77.0	258.6	10.1
S	87-1	–29.1, 26.2	6/6	333.6	–59.9	63	8.5				65.2	260.0	12.3

Appendix 2 Karoo dolerite dykes (D) and sills (S) (Continued)

	Site No.	Site Lat./Long.	N	D	I	k	α	Bedding dip/azimuth	Structurally corrected D	Structurally corrected I	Pole Lat.	Pole Long.	Pole α
D	2	–28.5, 28.6	8/6	331.5	–56.7	74	7.8				64.4	270.4	10.9
D	3	–28.5, 28.6	6/5	340.2	–45.6	346	4.1				72.5	298.6	5.0
S	4	–28.5, 29.6	6/6	154.2	+48.2	226	4.5				–67.5	111.4	5.4
S	5	–28.7, 31.8	31.8	6/6	158.4	+43.9	17	11, 06	152.0	+53.4	–65.6	102.3	16.5
S	6	–28.7, 31.8	31.8	6/5	170.7	+36.1	114	11, 06	167.9	+46.6	–79.3	123.4	7.8
D	7	–28.7, 31.8	31.8	6/6	129.9	+84.8	31	45, 180	174.7	+41.5	–83.2	166.1	22.7
Sabie River basalt													
	90-20	–27.5, 32.0	5/5	163.6	+57.1	40	12.3	10, 90	150.3	53.2	–62.1	93.2	14.0
Swaziland basalt dyke cutting uppermost Mbuluzi rhyolite													
	90.12	–26.2, 32.0	5/5	216.2	56.5	94	8.0	30, 74	157.4	70.1	–55.0	54.7	8.5

*Intersecting great circles

Lebombo Group Rhyolites

Site No.	Site Lat./ Long.	N	D	I	k	α	Bedding dip/azimuth	Structurally corrected D	Structurally corrected I	Pole Lat.	Pole Long.	Pole α
Jozini												
87-15	–27.8,32.2	4/4	196.1	47.1	201	6.5	20°,90	173.6	48.8	–83.5	101.6	6.6
18	–27.5,32.1	5/5	150.2	54.9	1838	1.8	10°,90	139.7	49.1	–54.5	108.3	2.2
19	–27.5,32.1	4/4	173.0	63.1	47	13.5	10°,90	155.0	60.3	–65.3	83.3	15.1
20	–27.5,32.1	6/6	178.9	50.5	671	2.6	10°,90	167.1	49.3	–80.0	102.9	2.8
21	–27.5,32.0	6/6	188.3	58.0	399	3.4	10°,90	172.2	58.0	–77.4	58.0	3.9
22	–27.5,32.0	5/5	165.3	56.4	336	4.2	10°,90	152.1	52.7	–65.2	101.5	5.5
88-2+90-18	–27.5,32.1	8/5	162.6	17.9	24	15.9	10°,90	159.8	14.7	–62.3	164.6	15.9
88-3+90-19	–27.5,32.1	8/4	333.0	–42.2	29	17.4	10°,90	325.8	–37.1	58.2	306.8	17.3
90-20*	–27.5,32.0	5/5	163.6	57.1	40	12.3	10°,90	150.3	53.2	–62.1	93.2	14.0
Mbuluzi												
88-7	–26.2, 32.0	4/3	326.6	–52.6	40	19.9	30°,80	302.6	–34.6	36.9	297.2	
8	–26.2, 32.0	4/4	06.6	–52.1	25	18.6	50°,54	287.3	–55.7	28.4	273.2	18.9
9	–26.2, 32.0	4/4	245.7	+36.0	8	35.0	60°,90	157.1	+68.8	–48.5	89.0	8.7
10	–26.2, 32.0	4/4	71.9	–18.6	42	14.3	90°,86	301.9	–66.8	40.1	258.3	8.0
							75°, 86	339.1	–76.0	50.3	226.5	8.0
90-11	–26.2, 32.1	5/5	07.7	–68.5	153	6.2	30°,74	298.8	–61.6	36.5	264.7	9.3
13	–26.2, 32.1	5/3	06.8	–66.0	201	5.4	30°,90	314.2	–54.6	48.8	274.4	7.2
14	–26.2, 32.1	5/5	14.6	–61.2	55	10.4	28°,90	326.9	–56.2	58.7	268.8	14.4
15	–26.2, 32.1	5/4	350.6	–47.4	45	13.8	32°,90	324.1	–34.4	56.3	302.4	12.1
16	–26.2, 32.1	5/5	359.6	–56.5	440	3.7	33°,80	315.3	–48.5	49.8	283.0	4.6
17	–26.2, 32.0	5/4	005.3	–60.5	10	30.5	15°,90	339.7	–58.5	66.7	254.3	45.4
Kruger												
88-11	–25.4, 32.0	4/4	176.8	–46.7	4	50.4	30°,85	205.7	–39.7	–30.6	241.6	50.7
12	–25.3, 31.9	5/5	07.7	–48.5	540	3.3	30°,85	334.0	–46.2	66.6	290.6	3.7
15	–23.8, 31.7	5	No significant grouping									
16	–23.8, 31.7	6/3	196.2	–19.0	9.2	43.2	15°,90	200.5	–14.3	–51.8	246.1	37.9
18	–23.8, 31.6	5/4	346.4	–55.8	73	10.8	15°,90	328.0	–49.9	61.3	283.7	14.0
19	–23.8, 31.6	a)5/3	346.8	–31.9	8	47.6	15°,90	338.6	–27.4	67.2	326.0	35.7
		b)5/2	153.8	–37.6	25	52.0	15°,90	165.8	–42.8	67.2		
90-8 –24.0,	–24.0, 31.7	6	No significant grouping									

*Sabie River basalt immediately below Jozini

Appendix 2 Karoo dolerite dykes (D) and sills (S) (Continued)

Mashikiri/Letaba, Kruger Park

Site No.	Site Lat./ Long.	N	D	I	k	α	Bedding dip/azimuth	Structurally corrected D	Structurally corrected I	Pole Lat.	Pole Long.	Pole α
Mashikiri												
88-13	–23.8, 31.6	5/4	348.8	–50.8	47	13.6	05/63	342.7	–51.9	72.5	267.9	16.6
14	–23.8, 31.6	4/4	341.6	–48.7	60	12.0	05/63	335.9	–49.2	67.7	280.0	13.1
90-1	–23.5, 31.5	7/4a*	347.3	–48.5	84	10.1	05/79	341.7	–48.1	72.7	278.6	9.1
		7/4b	149.8	+47.9	35	15.7	05/90	144.8	+46.0	–58.1	107.0	19.3
4	–23.5, 31.5	6/5	151.3	02.2	16	19.4	05/90	151.2	–0.4	–53.9	156.4	16.0
5	–22.5, 31.3	7	No Consistent Results									
Letaba												
90-6	–22.5, 31.3	6/6	155.6	45.2	46	10.0	05/90	151.3	+43.0	–63.7	110.1	11.8
E–W Dyke												
90-9	–25.0, 31.6	4/4	195.7	60.8	937	3.0	05/90	186.7	61.8	–71.2	016.3	5.1
							Excluding 90-9					
							Mean pole, N=5			–67.1	102.3	6.6
							Including Hooper *et al.* (1993)					
							Site KOL-18, N=6			–69.0	99.2	6.8

*Four samples from Site 90-1 have two distinct N & R components

Rooirand Dyke Swarm

Site No.	Site Lat./Long.	N	D	I	k	α	Bedding dip/azimuth	Structurally corrected D	Structurally corrected I	Pole Lat.	Pole Long.	Pole α
87-23	–27.5,31.9	4/4	211.8	23.5	42	14.3	20/90	201.9	32.6	–67.4	281.0	15.9
24	–27.5,31.9	7/7	178.4	44.3	9	21.0	20/90	160.2	40.5	–73.9	123.3	16.0
		7/4	203.0	54.6	283	5.5	20/90	172.9	57.5	–77.8	59.4	7.8
25	–27.5,31.9	4/4	152.1	–15.0	13	26.3	20/90	155.2	–9.8	–49.4	172.8	19.4
		4/3R*	345.1	–74.5	39	20.0	20/90	303.2	–61.9			
26	–27.5,31.9	4/4	146.8	11.2	64	11.6	20/90	145.5	+1.0	–47.3	155.2	5.8
		/4R*	09.6	–64.0	10	30.2	20/90	331.1	–60.4			
27	–27.5,31.8	4/4	25.0	–63.1	128	8.2	20/90	345.5	–65.5	+66.9	237.4	10.3
		28	–27.4,	–27.5,31.8	4/4	5.4	–65.2	21	20.3	20/90	326.3	–60.1
		4/3	25.9	–68.2	4713	1.8	20/90	333.7	–68.1	+58.7	244.9	2.0
29	–27.5,31.8	4/3	200.7	56.3	34	21.4	20/90	168.3	58.5	–75.2	71.4	25.1
88-6	–27.5,31.8	6/5	206.2	82.9	26	15.3	20/90	111.1	72.0	–34.1	69.7	29.5
							Excluding 88-6					
							Mean pole, N=8			–77.3	116.0	18.8

*Sites 25 and 26 have two distinct components, N(soft) and R

Bumbeni Complex, Natal

Site No.	Site Lat./Long.	N	D	I	k	α	Bedding dip/azimuth	Structurally corrected D	Structurally corrected I	Pole Lat.	Pole Long.	Pole α
87-8	–27.8, 32.3	7/7	314.4	–60.3	97	6.1	20,90	298.8	–44.0	+35.9	289.3	8.4
9	–27.8,32.3	7/7	303.1	–59.7	23	14.3	20,90	291.7	–41.8	+29.3	288.8	19.2
13	–27.8,32.3	9/7	273.8	–22.6	13	17.5	20,90	273.5	–02.6	+03.7	302.8	17.5
14	–27.8,32.2	6/6	198.0	–07.3	7	27.5	20,90	199.4	–00.8			
16	–27.7,32.3	6/6	108.3	50.0	241	4.3	05,90	106.7	45.2	–25.9	104.3	4.8
17	–27.7,32.3	6/6	111.9	54.1	26	13.3	05,90	109.6	49.4	–29.4	101.4	13.2
88-4	–27.7,32.3	6/4	282.5	–58.2	11	29.3	05,90	281.0	–53.3			
5	–27.9,32.3	8/3	119.9	–37.8	6	54.5	20,90	132.2	–54.1			
		8/2	150.2	+46.5	16	67.6	20,90	136.4	+34.4			
							Excluding 87-13					
							Mean pole, N = 4			30.2	285.9	6.0

Appendix 2 Karoo dolerite dykes (D) and sills (S) (Continued)

Cape Peninsula dolerites

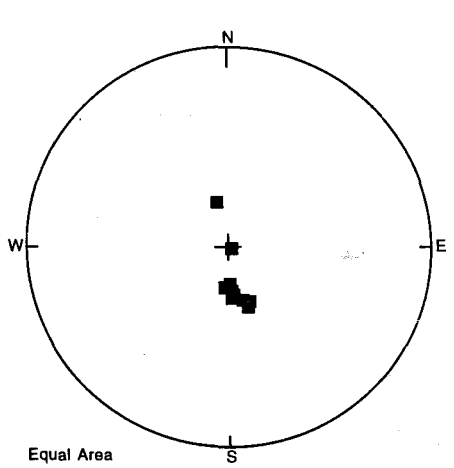
	Site						Bedding	Structurally corrected		Pole		
Site No.	Lat./Long.	N	D	I	k	α	dip/azimuth	D	I	Lat.	Long.	α
85-20	–34.2, 18.5	7/5	308.5	–48.6	563	3.2				–46.5	99.8	3.1
21	–34.0, 18.4	9/6	305.2	–33.6	84	7.3				–38.5	111.6	6.6
27	–32.0, 28.0	6/6	301.6	–50.1		13.2, 18.0*				–41.0	103.6	10.0
							MeanPole, N = 3			–42.2	105.2	9.1

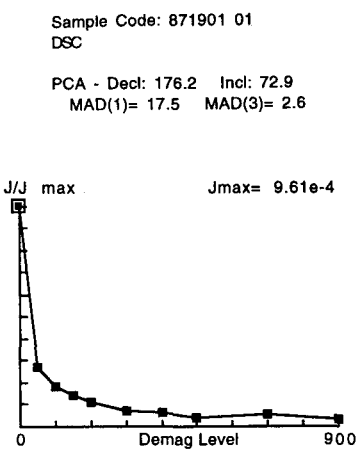
*intersecting great circles

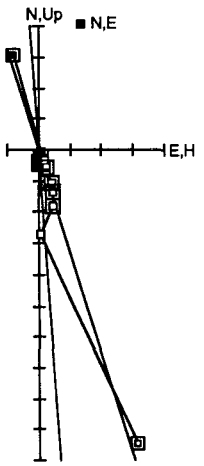
Appendix 2 Continued

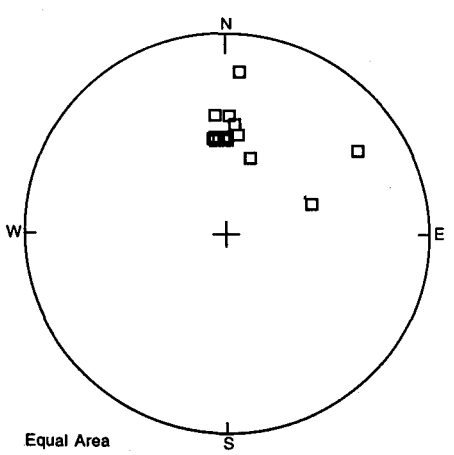
Representative demagnetization results: From left to right: Equal-area stereographic projection, normalized intensity versus AF; Zijderveld plot; PCA vector and MAD values shown beneath sample ID in upper center

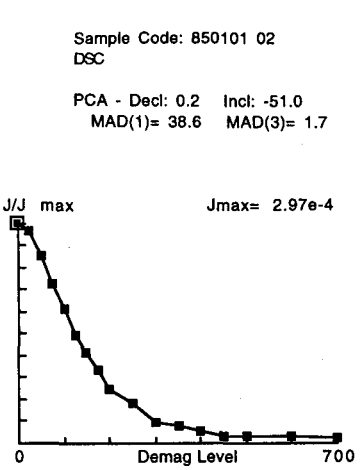
Core-sample	85-1-1-2	Dyke near Kimperley, CP; N
	85-19-6-1	Dolerite sill near Worcester, C.P; N
	87-6-4-1	Dolerite sill (?) near Tugela, Kwa-Zulu Natal; R
	87-19-1-1	Jozini rhyolite, Flow No. 4, Kwa-Zulu Natal; R

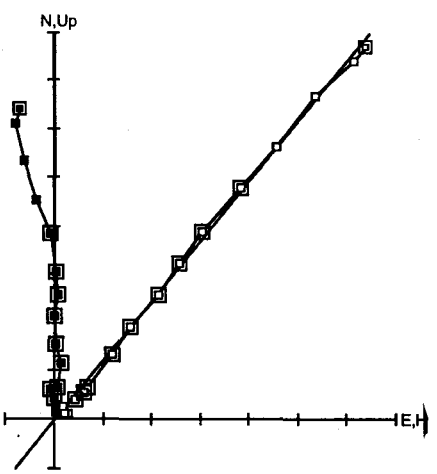












Appendix 2 (Continued)

